

29) A system of linear equations with fewer equations than unknowns is sometimes called an undetermined system. Suppose that such a system happens to be consistent. Explain why there must be an infinite number of solutions.

Soln: Since the number of equations is less than the number of unknowns, the corresponding coefficient matrix has lesser rows compared to the columns. Let, the given system be consistent. $\therefore$ By Theorem 2, the augmented form of the matrix ^{in echelon form} does not contain a single row of the form $[0 \ 0 \ \dots \ 0 \ b]$ with $b \neq 0$. $\therefore$ There is not a single zero row of the coefficient matrix in echelon form. $\therefore$ By defn, each row of the matrix has a single pivot position. From condition 2 of the definition of the echelon form of the matrix, no column has more than one pivot and therefore the no. of pivot columns of the matrix is equal to the no. of rows of the coefficient matrix. ~~Since the no. of~~
~~pivot columns~~ By defn, no. of basic variables = no. of rows of coeff matrix
 By defn, no. of free variables = (no. of cols - no. of rows) of coeff matrix which is non-zero.
 By Theorem 2, the system has infinitely many solutions.

30) Give an example of an inconsistent undetermined system of two equations in three unknowns.

Soln: Claim: $\begin{cases} x+y+z=2 \\ 2x+2y+2z=5 \end{cases}$ is such a system.

Proof: The augmented matrix is: $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 5 \end{bmatrix}$

$$\text{Row 2} = \text{Row 2} - 2 \times \text{Row 1}$$

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

By Theorem 2, the augmented matrix is inconsistent since it contains at least one row of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$.